Maintenance optimization with imperfect maintenance of a Gamma Process

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Abstract

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1. Introduction

2. Degradation process and maintenance model

2.1. Degradation process

We assume that the deterioration process is a Gamma process, $\{X_t, t \geq 0\}$, with shape parameter $a(t) = \alpha t^\beta$ and scale parameter b. The gamma process $\{X_t, t \geq 0\}$ is a continuous-time stochastic process such that:

- $X(0) = 0$ with probability 1
- $X(t) - X(s) \sim G(a(t) - a(s), b)$ for all $t > s \geq 0$, with density function $\forall x \in \mathbb{R}^+$:
$$f(x) = \frac{b^{a(t)-a(s)}}{\Gamma(a(t) - a(s))} x^{a(t)-a(s)-1} e^{-bx}$$
- $X(t)$ has independent increments

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Let $\{Y_t, t \geq 0\}$ denotes the stochastic process of the maintained system. We consider that the system is inspected at time $\tau, 2\tau, \ldots$. At each inspection ($t_j = j\tau$), a decision is taken in respect to the level of degradation:

- if $Y_{t_j} \geq M$ then a Corrective Maintenance (CM) is performed and the system is replaced by a new one (As Good As New);

- if $L \leq Y_{t_j} < M$ then a Preventive Maintenance (PM) is performed. We consider here an imperfect PM, and more precisely, an ARD_1 model, parameterized by ρ, i.e. the degradation level just after the inspection (at time t_j) is reduced by a proportional quantity of the degradation level just before the inspection: $Y_{t_j^+} = (1-\rho)Y_{t_j^-}$;

- if $Y_{t_j} < L$ no action is performed,

where L and M are fixed thresholds ($0 < L < M$).

Let $U_i = \mathbb{1}_{L \leq Y_{i\tau} < M}$ and

Between the inspection at time $(i-1)\tau$ and the inspection at time $i\tau$, the system is assumed to evolve according to i.i.d. copies $(X_t^{(i)})_{t\geq 0}$ of the gamma process $(X_t)_{t\geq 0}$ and we denote $W_i = X_{i\tau}^{(i)} - X_{(i-1)\tau}^{(i)}$ its increment between these two inspections.

Thus, before the first inspection, we have

$\forall t \in [0, \tau)$, $Y_t = X_t^{(1)}$

At the first inspection, we could have:

- if $Y_{\tau^-} < L$, no action is performed and then $Y_\tau = Y_{\tau^-} = X_{\tau^-}^{(1)}$.

- if $L \leq Y_{\tau^-} < M$, an imperfect PM (ARD_1) is performed and then the degradation level is reduced of $\rho X_\tau^{(1)}$ and thus $Y_\tau = (1-\rho)X_{\tau^-}^{(1)}$.

- if $Y_\tau^- > M$ then the system is replaced by a new one.

After the first inspection without replacement, we have $\forall t \in [\tau, 2\tau)$,

$$Y_t = Y_\tau + X_t^{(2)} - X_\tau^{(2)} = (1-\rho)^{U_1} X_\tau^{(1)} + X_t^{(2)} - X_\tau^{(2)}$$

At the second inspection at time 2τ, we have

- if $Y_{2\tau^-} < L$, no action is performed, and $Y_{2\tau} = Y_{2\tau^-} = (1-\rho)^{U_1} X_\tau^{(1)} + X_{2\tau}^{(2)} - X_\tau^{(2)}$.

- if $L \leq Y_{2\tau^-} < M$, an imperfect PM (ARD_1) is performed and then the degradation level is reduced of $\rho(X_{2\tau}^{(2)} - X_\tau^{(2)})$ and thus

$$Y_{2\tau} = Y_{2\tau^-} - \rho(X_{2\tau^-}^{(2)} - X_\tau^{(2)}) = (1-\rho)^{U_1} X_\tau^{(1)} + (1-\rho)(X_{2\tau^-}^{(2)} - X_\tau^{(2)})$$

- if $Y_{2\tau^-} > M$ then the system is replaced.

By induction, we can conclude that after i inspections without replacement, we have $\forall t \in [i\tau, (i+1)\tau)$

$$Y_t = Y_{i\tau} + (X_t^{(i+1)} - X_{i\tau}^{(i+1)})$$

and if we denote $i = \lfloor t/\tau \rfloor$, then we get

$$Y_t = \sum_{j=1}^{i} (1-\rho)^{1-U_j} (X_{j\tau}^{(j)} - X_{(j-1)\tau}^{(j)}) + X_t^{(i+1)} - X_{i\tau}^{(i+1)}$$

Finally, we obtain

$$\begin{aligned} Y_{(i+1)\tau} &= Y_{i\tau} + (1-\rho)^{U_{i+1}} (X_{(i+1)\tau}^{(i+1)} - X_{i\tau}^{(i+1)}) \\ &= \sum_{j=1}^{i+1} (1-\rho)^{U_j} (X_{j\tau}^{(j)} - X_{(j-1)\tau}^{(j)}) \end{aligned} \quad (1)$$

2.2. Maintenance Optimization

Let C_I, C_P anc C_C denote respectively the cost of an Inspection, a Preventive Maintenance and a Corrective Maintenance. As the corrective maintenances are perfect (replacement), we can compute the total cost on a renewal cycle $[0, T]$:

$$C(T) = C_C + [T/\tau]C_I + N_p C_p = C_C + [T/\tau]C_I + \sum_{i=1}^{T/\tau} \mathbb{1}_{L \leq Y_{i\tau} < M} C_p.$$

The length of a cycle is

$$T = \inf\{t > 0,\, Y_t \geq M\}$$

and the expectation of the length of a renewal cycle is:

$$E(T) = \int P(Y_t > M)dt$$

$E(C(T))/E(T)$

$$E(C(T) = C_c + E[T/\tau]C_i + E\sum_{i=1}^{T/\tau} P(M \leq Y_{i\tau} < L)C_P$$

$E(T) = \int P(Y_t > L)dt$

$F_T(t) = P$ $T = \inf\{t > 0, Y_t > L\}$ $Y_t = \sum_{i<t} G_i - \sum_{j<t/\tau} Z_i$ $Y^-_{i\tau} = G_{i-1} + Y^+_{(i-1)\tau}$

$Y^+_{i\tau} = (\rho)^{I_{M \leq Y_i \tau < L}} Y^-_{i\tau}$

2.3. Stationary law of the Maintained System

Let consider the Markov renewal cycle $[t^-_{i-1}, t^-_i]$, x and y respectively the deterioration level of maintained system at the beginning and the end of the cycle, the stationary law, π is the solution of :

$$\pi(y) = \int_0^{+\infty} \pi(x)p(y|x)\,dx \tag{2}$$

where $p(y|x)$ is the transition law from the state x to the state y. This transition law is equal to

$$p(y|x) = f(y-x)\mathbb{1}_{x<L} + f(y-(1-\rho)x)\mathbb{1}_{L\leq x<M} + f(y)\mathbb{1}_{x>M} \tag{3}$$

where f is the density function of gamma distribution :

$$f(x) = \frac{\beta^{\alpha(t_i^\beta - t_{i-1}^\beta)}}{\Gamma(\alpha(t_i^\beta - t_{i-1}^\beta))} x^{\alpha(t_i^\beta - t_{i-1}^\beta)-1} e^{-bx}$$

Thus, we have

$$\pi(y) = \int_0^L \pi(x)f(y-x)\,dx + \int_L^M \pi(x)f(y-(1-\rho)x)\,dx + f(y)\int_M^{+\infty} \pi(x)\,dx \tag{4}$$

We can solve this last equation by fixed-point iteration algorithm by considering that $\pi(x)$ is the solution of $\pi(y) = g(\pi(y))$, where g is a continuous function. Thus, for all y, we approximate eq. (5) at iteration k by

$$w_k(y) = \int_0^L w_{k-1}(x)f(y-x)\,dx + \int_L^M w_{k-1}(x)f(y-(1-\rho)x)\,dx + f(y)\int_M^{+\infty} w_{k-1}(x)\,dx$$

Pour moi, il faut plutôt écrire

Thus, we have

$$\pi(y) = \int_0^{\min(L,y)} \pi(x) f(y-x)\, dx + \int_{\min(\frac{y}{1-\rho}, L)}^{\min(\frac{y}{1-\rho}, M)} \pi(x) f(y-(1-\rho)x)\, dx + f(y) \int_M^{+\infty} \pi(x)\, dx \tag{5}$$

We can solve this last equation by fixed-point iteration algorithm by considering that $\pi(x)$ is the solution of $\pi(y) = g(\pi(y))$, where g is a continuous function. Thus, for all y, we approximate eq. (5) at iteration k by

$$w_k(y) = \int_0^{\min(L,y)} w_{k-1}(x) f(y-x)\, dx + \int_{\min(\frac{y}{1-\rho}, L)}^{\min(\frac{y}{1-\rho}, M)} w_{k-1}(x) f(y-(1-\rho)x)\, dx + f(y) \int_M^{+\infty} w_{k-1}(x)\, dx$$

We set the initial value of w_k as $w_1(x) = x\exp(-x)$? Impossible pour moi d'initialiser à 1 car la dernière intégrale de l'équation va diverger... Pourquoi pas $w_1(x) = x\exp(-x)$?

3. Estimation aux instants

Normalement on observe les $Y_{t_1}^-, Y_{t_2}^-, Y_{t_3}^- \ldots\ldots Y_{t_n}^-$ Le nombre d'instant d'inspection Deux possibilités: a) Des données aux instants d'inspections (savoir ou non si maintenance ou pas) b) Des données aux instants de maintenance préventive.

SI ρ aléatoire il faut estimer les paramètres de la loi de ρ

Choix d'observations a)$Y_{t_1}^-, Y_{t_2}^-, Y_{t_3}^-, \ldots, Y_{t_n}^-, Y_{t_1}^+, Y_{t_2}^+, Y_{t_3}^+, \ldots, Y_{t_n}^+$ On aura donc des observations de ρ en utilisant la fraction et les increments $Y_{t_i}^- - Y_{t_{i-1}^+}$ permettent d'estimer les paramètres du processus gamma

b) $Y_{t_1}^-, Y_{t_2}^-, Y_{t_3}^-, \ldots, Y_{t_n}^-$ Plus difficile pour estimer les paramètres de la loi de ρ